

PUSHKAR DUBE

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EDUCATION

- **B.Tech in Electronics and Telecommunication Engineering**, Veermata Jijabai Technological Institute (VJTI)
– Aug 2024 – Present | Current CGPA: 7.72 / 10

EXPERIENCE

- **UWB Ranging Security Research** IMEC-COSIC, KU LEUVEN, BELGIUM *Jun 2026 – Present*
UWB Security — IEEE 802.15.4z HRP — Embedded Firmware — Wireless Attacks — Reverse Engineering — Zephyr RTOS
 - Implemented SS/DS-TWR and STS-secured (IEEE 802.15.4z) ranging firmware on the Qorvo DWM3001CDK (DW3110), with antenna-delay calibration, delayed-TX scheduling, and CIA timestamp gating.
 - Reproduced and demonstrated UWB attacks on real hardware: Cicada pulse-train, Ghost Peak distance reduction (approximately 5 m against an STS-secured link), and UWBAD selective/flood jamming against Qorvo CLI firmware and the Apple NI iPhone link on a bare chip.
 - Reverse-engineered live commercial FiRa and Apple NI traffic, decoding 802.15.4z MAC framing, slot grid, and STS config from passive captures; built a FAST-based attack success-rate pipeline with Clopper-Pearson intervals.
- **High-Precision UWB Localization System** KRISHNA DEFENCE AND ALLIED INDUSTRIES LTD. *Jan 2026 – Apr 2026*
UWB Localization — Embedded Systems — Real-Time Systems — Signal Processing — DS-TWR — TDoA
 - Designed and deployed a centimeter-scale localization system using Qorvo DWM3001 modules (IEEE 802.15.4z HRP UWB).
 - Achieved approximately 2 cm ranging accuracy using Double-Sided Two-Way Ranging (DS-TWR).
 - Developing scalable TDoA infrastructure with multi-anchor synchronization for ship dry-docking applications.
 - Implemented custom firmware for timestamp acquisition, antenna delay calibration, and clock drift compensation.
 - Conducted structured error analysis addressing clock skew, synchronization drift, timestamp quantization, and multipath effects.
- **GantryMate-200 - Precision Industrial Automation** ELECTROLYTE SOLUTIONS *Jun 2025 – Sep 2025*
Robotics — Motion Control — Mechanical Design — Mechatronics — Gantry
 - Designed and built a 3-axis automated gantry for precision screwing of approximately 280 miniature (1 mm) screws, at approximately 0.05 mm repeatable positioning.
 - Developed embedded firmware (ESP32, ESP-IDF) for coordinated motion control and homing routines.
 - Integrated NEMA17 stepper motors with DRV8825 and TMC2208 drivers using microstepping.
 - Designed mechanical CAD and a custom PCB for motor-driver integration and power distribution.

PROJECTS

- **PlatyPulse – ESP32-S3 Multi-Tool Signal Device (Mentor)** *Jul 2026 – Present*
Embedded Systems — ESP-IDF — FreeRTOS — RF/IR/NFC — PCB Design — Code
 - Built a Flipper Zero-inspired ESP32-S3 multi-tool device on ESP-IDF/FreeRTOS, with independent tasks driving CC1101 sub-GHz RF capture/replay (300–928 MHz), IR transceiver (NEC/Sony/RC5 via RMT), PN532-based NFC/RFID read, and OLED/button UI control.
 - Completed full KiCad schematic and PCB layout with a custom footprint library and routed power/SPI/UART nets, carrying the board through DRC ahead of fabrication.
- **Custom ESP32-Pico-D4 Smartwatch** *Feb 2026 – May 2026*
Embedded Systems — PCB Design — IoT — Low-Power Systems — Code
 - Designed full schematic and PCB layout (KiCad) integrating ESP32, MPU6050 IMU, OLED display, and battery management circuitry.
 - Implemented low-power firmware for sensor acquisition and wireless communication, with a regulated, protected power architecture.
- **Wireless Multi-Node Embedded System (ESP32-C3)** *Nov 2025 – Dec 2025*
Distributed Systems — IoT — Wireless Communication — Embedded Firmware — Code
 - Designed a custom PCB integrating power regulation (LM2596), sensor interfaces, and motor drivers.
 - Implemented ESP-NOW communication with CRC-8 integrity validation, OTA firmware updates, and a SoftAP configuration interface.
- **Maze Solving Robot** *Oct 2025 – Nov 2025*
Robotics — Path Planning — Control Systems — State Estimation — Code
 - Designed and built an autonomous robot capable of solving unknown mazes using real-time decision making.
 - Applied low-pass filtering and Kalman filtering for sensor noise reduction and improved state estimation.
 - Implemented dynamic duty cycle control for stable motion and adaptive speed regulation.

TECHNICAL SKILLS

- **Languages:** C, C++, Python
- **Embedded:** ESP32 (ESP-IDF), Zephyr RTOS, nRF Connect SDK (NCS)

- **Wireless & Communication:** Ultra-Wideband (UWB), ESP-NOW
- **Robotics & Simulation:** ROS2, Gazebo, Mujoco
- **Design & CAD:** KiCad (PCB Design), Onshape, Autodesk Fusion
- **Development & Tools:** Linux, Git, GNU Radio

ACTIVITIES

Society of Robotics and Automation (SRA), VJTI

SPONSORSHIP HEAD

- Contributed to institutional robotics projects focused on embedded firmware and professional PCB design.
- Conducted workshops for 400+ first-year students on:
 - ESP32-based Line Following Robot – Wall-E
 - Image Processing Fundamentals – Pixels
 - ROS2-based 3DOF Manipulator – MARIO